AdifySkills

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📘 CLASS NOTES

🧩 Module 2 · Core Nodes & Data Mastery

DAY 04 of 25

# Expressions & Data Transformation

"{{ }} — n8n ki asli superpower."

⚡ AAJ HUM YEH BANAYENGE

## Messy Lead List Cleaner

Gande data ko saaf karta hai: naam proper case mein, phone +91 format mein, aur galat emails alag. Saaf leads ek tab mein, galat wale "Review" tab mein.

Edit Fields | Date & Time | Expressions

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

# 🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

**💡 Yeh lesson kyun zaroori hai?**

Expressions se workflow "smart" banta hai — har item ke data ke hisaab se alag message, alag date, alag value.

## 🎯 Aaj ke baad aap yeh kar paoge

**01 Expression editor & $json, $input, $node**

🌍 Form ka "Naam" box — har form mein alag value.
⚙️ {{ $json.email }} → current item ka email.

**02 Referencing previous nodes: $("Node").item**

🌍 Pichhle page par wapas jaake dekhna kisne bheja tha.
⚙️ {{ $("Webhook").item.json.body.phone }}

**03 String, number, array & date functions**

🌍 Calculator: plus, minus, uppercase, ginti.
⚙️ {{ $json.price * 1.18 }}, {{ $json.tags.join(", ") }}, {{ $json.name.toUpperCase() }}

**04 Luxon for dates ($now, $today, formats, timezones)**

🌍 Calendar mein "aaj se 7 din baad" dekhna.
⚙️ {{ $now.plus({days: 7}).toFormat("dd LLL yyyy") }} → "08 Dec 2026"

**05 Edit Fields (Set) node — mapping & renaming**

🌍 Form ke columns ka naam badalna: "Mob No" → "phone".
⚙️ Edit Fields: phone = {{ $json["Mob No"] }}, source = "website"

**06 Drag-and-drop mapping best practices**

🌍 Puzzle piece ko sahi jagah drag karna.
⚙️ Input panel se field ko parameter box mein drag → expression khud ban jaata hai.

## ⏱ Class ka plan (2 ghante)

| 10m | 35m | 45m | 20m | 10m |
|---|---|---|---|---|

**0:00-0:10 Recap + Quiz results**

*🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

**0:10-0:45 Concept: Expressions**

20 must-know expressions

*🪄 Mad-libs game jaisa — sentence mein khaali jagah hai, har baar alag word bharo, alag sentence ban jaata hai.*

**0:45-1:30 Live Build**

Lead data cleaner — name/phone/email normalize

*🪄 Aaj ki kahani yaad hai — 'Shaadi ke card par naam bharna'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

**1:30-1:50 Expression Gym**

10 rapid challenges

*🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

**1:50-2:00 Q&A + Homework**

*🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

# 🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

### {{ }} Expression

Double curly braces ke andar JavaScript chalti hai.

🪄 **Analogy:** Mail-merge ke placeholders

CONCEPT 02

### $json

Current item ka data. $json.email = is item ka email.

🪄 **Analogy:** "Yeh wala form"

CONCEPT 03

### $("Node Name")

Kisi bhi pichhle node ka data access.

🪄 **Analogy:** Pichhle page par wapas dekhna

CONCEPT 04

### Luxon ($now, $today)

Dates ke liye: $now.plus({days:3}).toFormat("dd LLL")

🪄 **Analogy:** Calendar calculator

### 🔑 Aaj ka main idea

Expression {{ }} ek "khaali jagah" hai jo har item ke data se bhar jaati hai. $json = current item, $("Node") = kisi pichhle node ka data, $now = abhi ka time. Isse ek hi workflow har customer ke liye alag message bana deta hai.

- Bank SMS: "Dear {{name}}, ₹{{amount}} debited"
- School report card: "{{student}} ne {{marks}} laaye"
- Birthday wish: "Happy birthday {{name}}, aaj aap {{age}} saal ke ho gaye!"

### 📖 Ek kahani se samjho: Shaadi ke card par naam bharna

1. 💌 **Ek hi card design** = Template (message)
2. ✍️ **Har card par alag naam** = {{ $json.name }}
3. 📅 **Date likhna** = {{ $now.toFormat() }}
4. 🔤 **Naam ka spelling theek** = .toTitleCase()
5. 📮 **100 card ready** = 100 personalised items

🪄 Ek line mein: “Mad-libs game jaisa — sentence mein khaali jagah hai, har baar alag word bharo, alag sentence ban jaata hai.”

# 🗺️ Aaj ka workflow: Messy Lead List Cleaner

Data left se right chalta hai. Har dabba ek n8n node hai.



**💰 Isse kitna kama sakte ho?**

Agencies, telecallers (lead list cleaning) — aam taur par **₹3,000 – ₹10,000 per project**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

## 🛠 Step-by-step build — n8n kholo aur saath chalo

1. ⚡ **Manual Trigger**
   Test
2. 👨‍💻 **Code**
   Sample messy leads (5 items) generate
3. ✏️ **Edit Fields (Set)**
   name → toTitleCase, phone → +91 format, email → lowercase trim
4. 🔀 **IF**
   Email valid hai? {{ $json.email.isEmail() }}
5. 📊 **Google Sheets**
   Valid → "Clean Leads", invalid → "Review" sheet

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

**🔑 Setup (import ke baad)**

- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Sheet mein 2 tabs banao: "Clean Leads" aur "Review" (headers: name, phone, email, email_valid)

**✅ Test kaise karein**

1. "Test workflow" dabao
2. "Clean Fields" ka output dekho — before/after compare karo
3. IF ke true/false dono branches mein kitne items gaye?

## ⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

**😩 Pehle (manual)**

| | |
|---|---|
| Har lead ka naam/phone manually theek karna | 30 min |
| Invalid emails dhoondhna | 15 min |
| | 45 min |

➜

**⚡ Ab (n8n)**

| | |
|---|---|
| Expressions se sab auto-clean | |
| Invalid alag sheet mein | |
| | 0 min |

⏱ 100 leads par 45 min → 0, aur data hamesha clean.

## ⌨️ Aaj ka key snippet

**Copy karke n8n mein paste karo** 📋 Select karke copy karo

```
{{ "+91" + $json.phone.replace(/\D/g, "").slice(-10) }}
```

## 📦 Sample data — input se output tak

**📥 Input** 📋 Select karke copy karo

```
{
  "name": "  rAHUL sharma ",
  "phone": "98765-43210",
  "email": " Rahul@GMAIL.com "
}
```

**📤 Output** 📋 Select karke copy karo

```
{
  "name": "Rahul Sharma",
  "phone": "+919876543210",
  "email": "rahul@gmail.com",
  "valid": true
}
```

# 💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

**💬 Build it with ChatGPT / Claude** 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Gande data ko saaf karta hai: naam proper case mein, phone +91 format mein, aur galat emails alag. Saaf leads ek tab mein, galat wale "Review" tab mein.

Flow: Manual Trigger (Test run) → Code (Sample messy leads) → Edit Fields (Expressions se clean) → IF (Email valid?) → Google Sheets (Clean / Review)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

**💬 Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Messy Lead List Cleaner" (nodes: Run Manually, Sample Messy Leads, Clean Fields, Email Valid?, Save Clean Leads, Save for Review).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

## 📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo ( `Ctrl` + `A` , `Ctrl` + `C` ).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

**Workflow JSON**
6 nodes · ready to import
`adifyskills.com/workflows/day-04-messy-lead-cleaner.json`

**Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/4`

# 🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

| Agar yeh ho raha hai… | To yeh karo |
|---|---|
| **Node par laal ⚠️ triangle?** | Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo. |
| **"Sheet not found" / column khaali?** | Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye. |

### ⚠️ Yeh galtiyan sabse common hain

- ! Expression mode (fx) on karna bhool jaana — text literally chala jaata hai
- ! Node ka naam badalne par $("Old Name") toot jaana
- ! Null field par .toLowerCase() → error (use ?. ya default)

### ⭐ Pro tips — time bachane wali tricks

- ★ Drag-and-drop mapping se expressions automatic bante hain
- ★ Expression editor mein live preview hamesha dekho
- ★ $json.field ?? "default" se null-safe raho

**Phir bhi atke? Koi baat nahi 🙂**
Screenshot + error message ke saath WhatsApp par poochho →

# 🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

### 🛠 Live lab

Messy lead list ko clean karna: proper case names, +91 phone format, invalid emails flag.

### 📝 Homework

Expressions cheat sheet se 15 expressions practise karke apni notebook workflow banao.

- ☐ Homework ho gaya

**BEGINNER**
Greeting: "Good morning, {{name}}"

**INTERMEDIATE**
Age calculate from DOB

**ADVANCED**
Indian festival ke hisaab se dynamic subject line

### 🧩 Aaj ka challenge — pehle khud try karo

**🎯 Email ka domain (gmail.com, yahoo.com) alag column mein nikaalo.**

💡 **Hint 1**
n8n mein strings ke liye ready-made functions hain — email ke liye bhi.

💡 **Hint 2**
Expression: {{ $json.email.extractDomain() }} try karo.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

## 🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

**Q1. Current item ka data kaise access karte hain?**

(A) $item
(B) $json
(C) $data
(D) $this

**Q2. 3 din baad ki date?**

(A) date+3
(B) $now + 3
(C) $now.plus({days: 3})
(D) $today(3)

**Q3. Expression kis syntax mein likhte hain?**

(A) <% %>
(B) (( ))
(C) [[ ]]
(D) {{ }}

**Q4. Kisi pichhle node "Webhook" ka data?**

(A) $("Webhook").item.json
(B) $prev
(C) $webhook
(D) $node.Webhook

## 📖 Aaj ke naye shabd

**Workflow**
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

**Node**
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana...).

**Trigger**
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

**JSON**
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

**Expression**
{{ }} ke andar likha chhota code jo data ko jagah par bhar deta hai, jaise {{ $json.name }}.

**Item**
Data ki ek row/record. n8n mein data items ki list ki tarah flow karta hai.

**Merge**

Do raaston ka data wapas ek jagah jodna.

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## ✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 5

**Logic & Flow Control: IF, Switch, Merge, Loop**

"Workflows jo khud decision lete hain."

☐ **Day 4 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

# 🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

**Q1 → B**
**$json**

**Q2 → C**
**$now.plus({days: 3})**
Luxon syntax.

**Q3 → D**
**{{ }}**

**Q4 → A**
**$("Webhook").item.json**

**✅ Challenge ka solution**

Clean Fields mein naya field: domain = {{ $json.email.trim().toLowerCase().extractDomain() }}

## 📚 Official docs

- Expressions ↗
- Built-in methods ↗
- Luxon dates ↗
- Edit Fields (Set) ↗

## ▶ Videos

- n8n expressions tutorial ↗
- n8n luxon date expressions ↗

**Doubt poochho**
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**Day 4 lesson page**
Quiz, copy buttons, videos
adifyskills.com/day/4

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